

Fuzzy Logic and Fuzzy Sets

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

Introduction

Crisp Set and Fuzzy Set

Fuzzy Logic vs Probability

Representation of Fuzzy Set

Linguistic Variables

Hedges

Operations of Fuzzy Sets

Introduction

Orange, Red, or Blue?



What is fuzzy logic?

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- ✓ logic to describe fuzziness

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Sunway is **quite a short** distance from Puchong.

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For example, the boundary is 180 cm. Samuel is 179 cm, therefore he is a short guy.

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Is Samuel really a short guy, or an arbitrary line has been drawn?

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Fuzzy logic makes it possible to avoid this situation.

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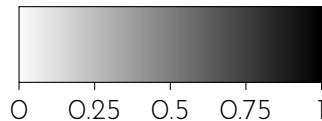
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- ▶ Zadeh extended the work into a formal system of mathematical logic.

Boolean and fuzzy logic

Range of logical values



Boolean logic



Fuzzy/multi-valued logic

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 x either belongs to the set A ($x \in A$),
or x does not belong to the set A ($x \notin A$).
- ▶ Characteristic function of crisp set A of universe X

$$f_A(x) : X \rightarrow 0, 1$$

where

$$f_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A \end{cases}$$

Example 1

From the following list, select the set each item belongs to.

Monday	butter	coffee	Thursday	keyboard
Friday	laptop	cat	Saturday	Sunday

Set of **NOT** days of week

Set of **days** of week

Example 2

day of week	Set of weekends	Set of not weekends
Monday		
Tuesday		
Wednesday		
Thursday		
Friday		
Saturday		
Sunday		

Fuzzy set

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- ▶ The value of a membership function represents the degree of membership, or called the membership value of x in set A .
- ▶ More typically represents the human way of thinking.

Crisp set and fuzzy set

Set of tall men

Name	Height (cm)	Degree of membership	
		Crisp	Fuzzy
Chris	208		
Mark	205		
John	198		
Tom	181		
David	179		
Mike	172		
Bob	167		
Steven	158		
Bill	155		
Peter	152		

Crisp set and fuzzy set

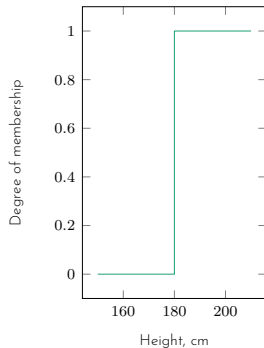
Set of tall men

Name	Height (cm)	Degree of membership	
		Crisp	Fuzzy
Chris	208	1	
Mark	205	1	
John	198	1	
Tom	181	1	
David	179	0	
Mike	172	0	
Bob	167	0	
Steven	158	0	
Bill	155	0	
Peter	152	0	

Crisp set and fuzzy set

Set of tall men

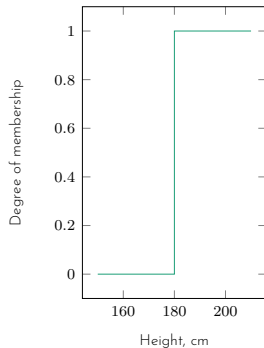
Name	Height (cm)	Degree of membership	
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Chris	208	1	
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Tom	181	1	
David	179	0	
Mike	172	0	
Bob	167	0	
Steven	158	0	
Bill	155	0	
Peter	152	0	



Crisp set and fuzzy set

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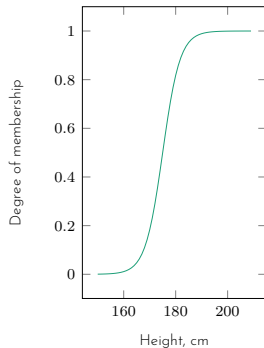
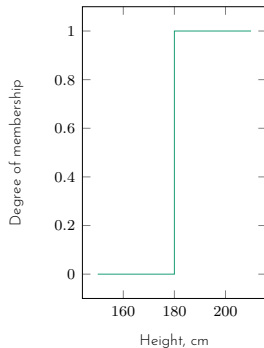
Name	Height (cm)	Degree of membership	
		Crisp	Fuzzy
Chris	208	1	1.00
Mark	205	1	1.00
John	198	1	0.98
Tom	181	1	0.82
David	179	0	0.78
Mike	172	0	0.24
Bob	167	0	0.15
Steven	158	0	0.06
Bill	155	0	0.01
Peter	152	0	0.00



Crisp set and fuzzy set

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Chris	208	1	1.00
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Crisp set and fuzzy set

Set of weekends (truth values for 'weekend-ness')

	Weekends	
	Crisp	Fuzzy
Monday		
Tuesday		
Wednesday		
Thursday		
Friday		
Saturday		
Sunday		

Crisp set and fuzzy set

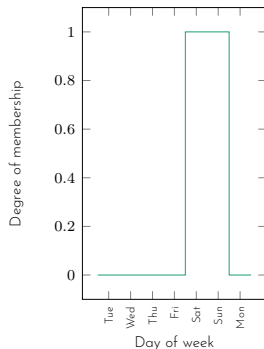
Set of weekends (truth values for 'weekend-ness')

	Weekends	
	Crisp	Fuzzy
Monday	0	
Tuesday	0	
Wednesday	0	
Thursday	0	
Friday	0	
Saturday	1	
Sunday	1	

Crisp set and fuzzy set

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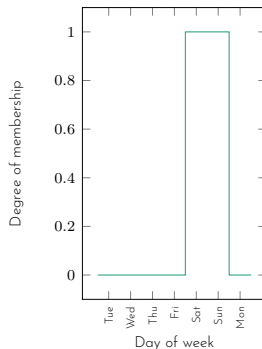
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Monday	0	
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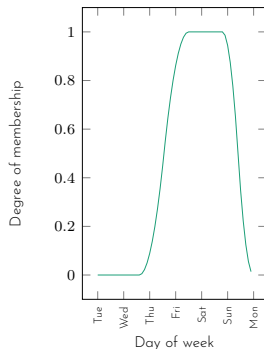
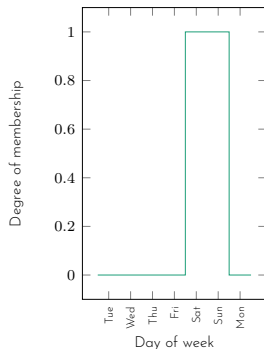
	Weekends	
	Crisp	Fuzzy
Monday	0	0.00
Tuesday	0	0.00
Wednesday	0	0.00
Thursday	0	0.20
Friday	0	0.85
Saturday	1	1.00
Sunday	1	0.98



Crisp set and fuzzy set

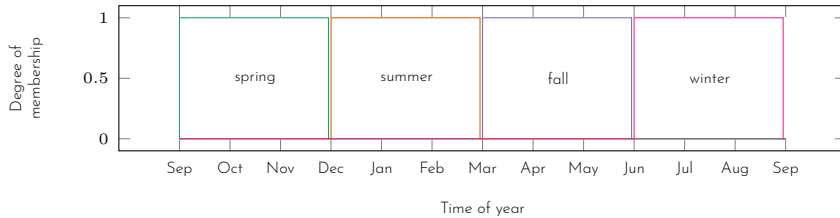
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Friday	0	0.85
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Sunday	1	0.98



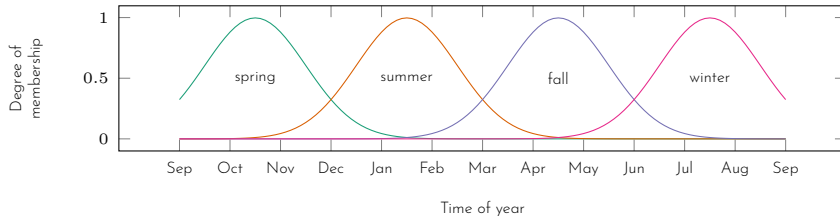
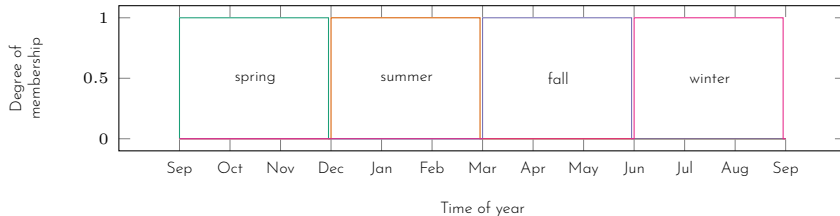
Crisp set and fuzzy set

Seasons of the year in southern hemisphere



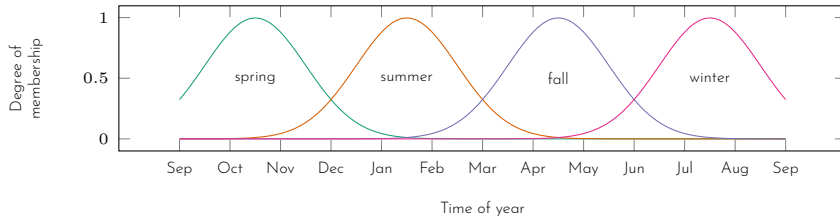
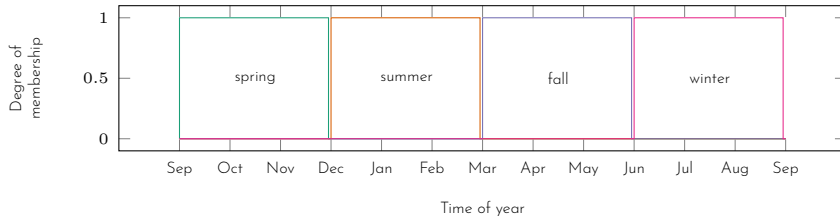
Crisp set and fuzzy set

Seasons of the year in southern hemisphere



Crisp set and fuzzy set

Seasons of the year in southern hemisphere



Is crisp set or fuzzy set a better representation in this case?

Fuzzy logic $\neq$ Probability

Fuzzy logic and probability

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Probability

Fuzzy Logic

Fuzzy logic $\neq$ Probability

Probability

represents the certainty of
an event happening

Fuzzy Logic

not representing the (un)certainty

Fuzzy logic $\neq$ Probability

Probability	Fuzzy Logic
represents the certainty of an event happening	not representing the (un)certainty
based on closed world model (everything is known)	not based on closed world model (there are things we do not know)

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based on frequency and requires independence of variables	does not require statistical independence of variables

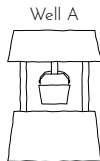
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Probability	Fuzzy Logic
represents the certainty of an event happening	not representing the (un)certainty
based on closed world model (everything is known)	not based on closed world model (there are things we do not know)
based on frequency and requires independence of variables	does not require statistical independence of variables
goes away once the event is observed	membership values persist after observation

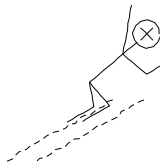
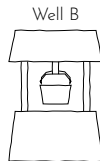
Fuzzy logic and probability

A man is lost in the desert. In the distance he can see two wells but he only has the strength to get to one.

Well A has a 0.9 probability of drinkable water

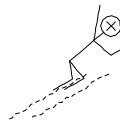
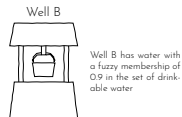


Well B has water with a fuzzy membership of 0.9 in the set of drinkable water



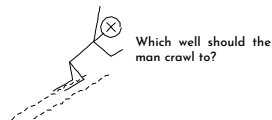
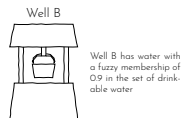
Which well should the man crawl to?

Fuzzy logic and probability



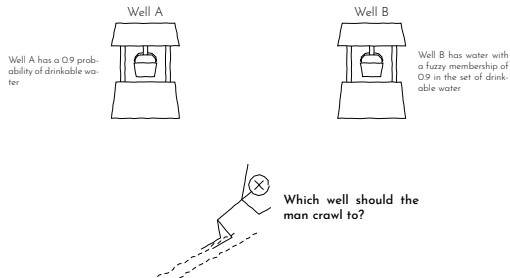
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Fuzzy logic and probability



Well A

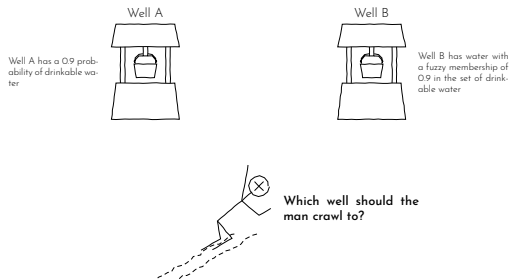
Fuzzy logic and probability



Well A

- ▶ The probabilistic information means that out of 10 wells, 9 will have suitable drinking water but 1 well will be poisonous.

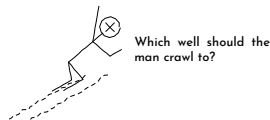
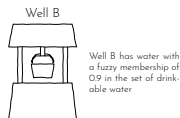
Fuzzy logic and probability



Well A

- ▶ The probabilistic information means that out of 10 wells, 9 will have suitable drinking water but 1 well will be poisonous.
- ▶ This one could be the poisonous one.

Fuzzy logic and probability

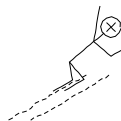
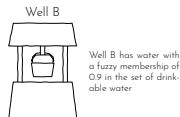


Well A

- ▶ The probabilistic information means that out of 10 wells, 9 will have suitable drinking water but 1 well will be poisonous.
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Well B

Fuzzy logic and probability



Which well should the man crawl to?

Well A

- ▶ The probabilistic information means that out of 10 wells, 9 will have suitable drinking water but 1 well will be poisonous.
- ▶ This one could be the poisonous one.

Well B

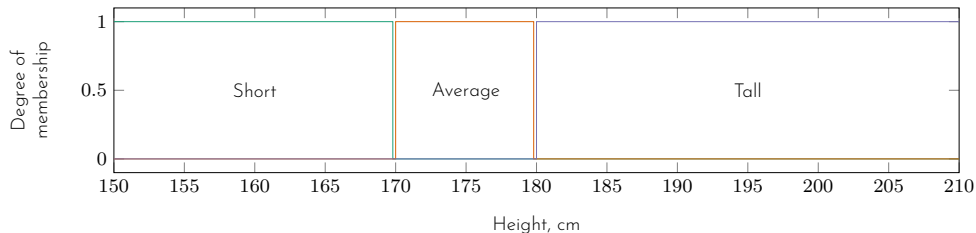
- ▶ The water quality is not perfect, but it is certain that the water is drinkable.

Representation of elements in a given set

Consider the crisp set and fuzzy set for the height of men.

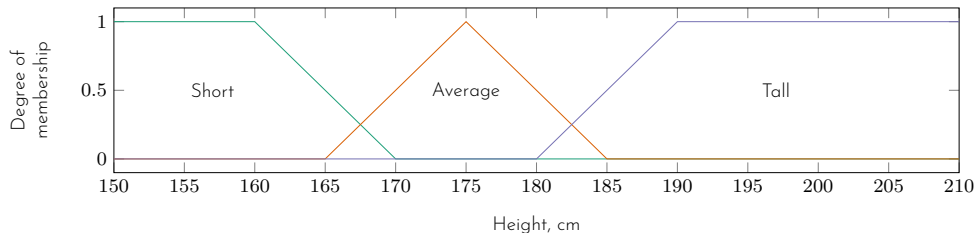
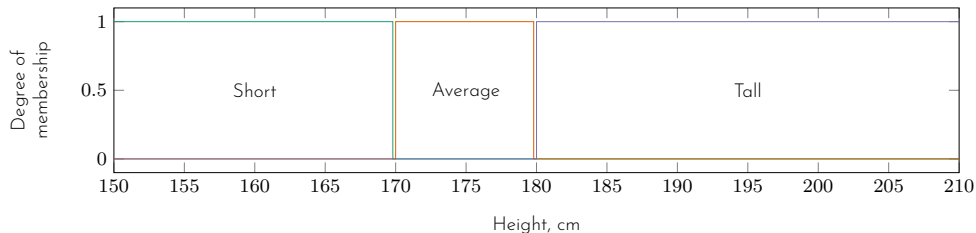
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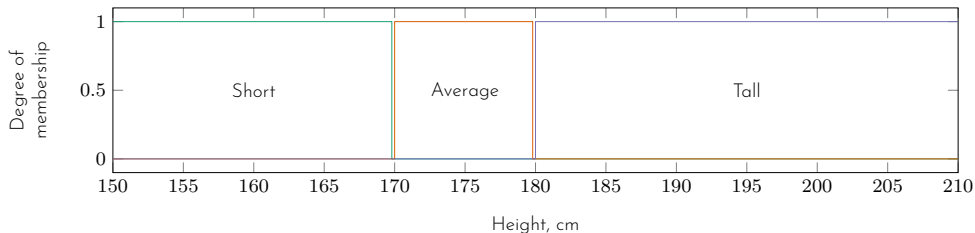


Representation of elements in a given set

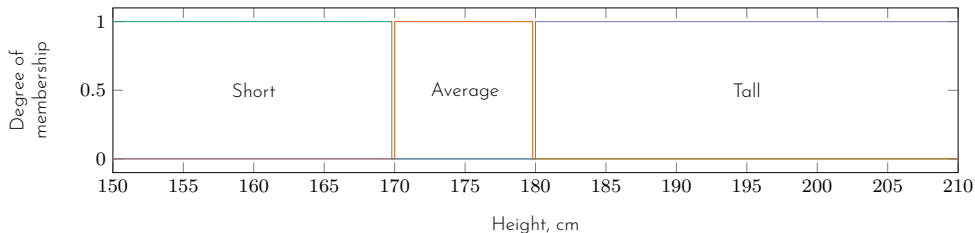
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Representation of elements in a given set



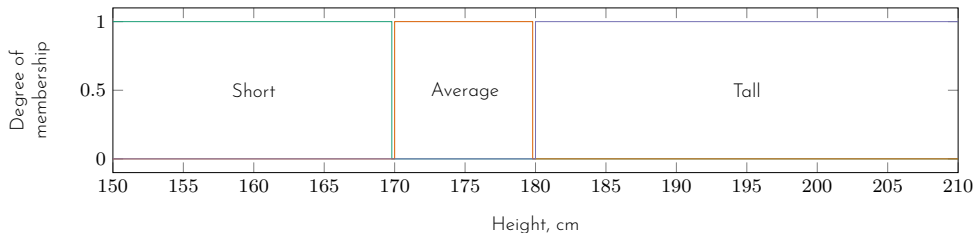
Representation of elements in a given set



Consider the universe X of 5 elements,

$$X = \{155, 162, 172, 178, 182\}$$

Representation of elements in a given set

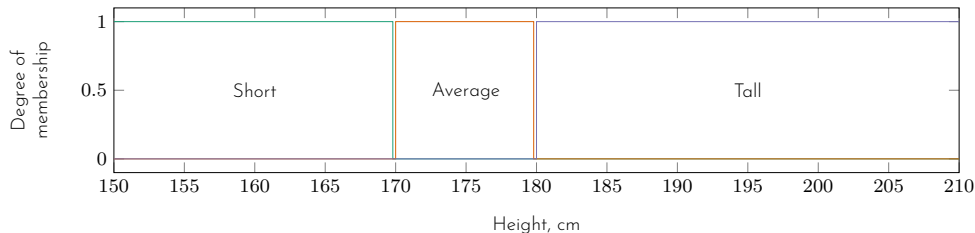


Consider the universe X of 5 elements,

$$X = \{155, 162, 172, 178, 182\}$$

How do we represent the crisp subset of short, average, and tall men in the universe X separately?

Representation of elements in a given set



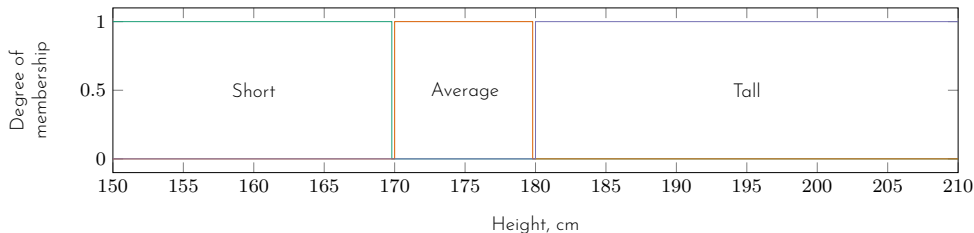
Consider the universe X of 5 elements,

$$X = \{155, 162, 172, 178, 182\}$$

How do we represent the crisp subset of short, average, and tall men in the universe X separately?

$$\text{short men} = \{(155, 1), (162, 1), (172, 0), (178, 0), (182, 0)\}$$

Representation of elements in a given set



Consider the universe X of 5 elements,

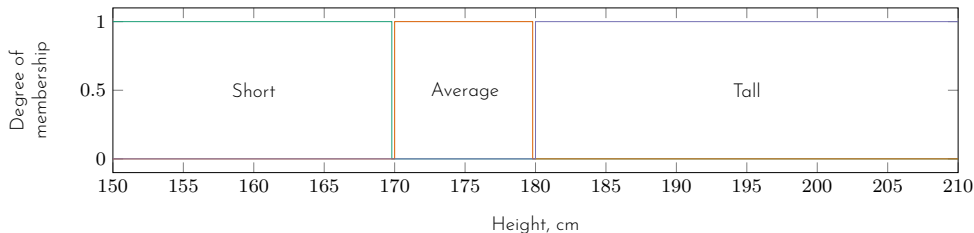
$$X = \{155, 162, 172, 178, 182\}$$

How do we represent the crisp subset of short, average, and tall men in the universe X separately?

$$\text{short men} = \{(155, 1), (162, 1), (172, 0), (178, 0), (182, 0)\}$$

$$\text{average men} = \{(155, 0), (162, 0), (172, 1), (178, 1), (182, 0)\}$$

Representation of elements in a given set



Consider the universe X of 5 elements,

$$X = \{155, 162, 172, 178, 182\}$$

How do we represent the crisp subset of short, average, and tall men in the universe X separately?

$$\text{short men} = \{(155, 1), (162, 1), (172, 0), (178, 0), (182, 0)\}$$

$$\text{average men} = \{(155, 0), (162, 0), (172, 1), (178, 1), (182, 0)\}$$

$$\text{tall men} = \{(155, 0), (162, 0), (172, 0), (178, 0), (182, 1)\}$$

Representation of elements in a given set

Consider the universe X with 5 elements,

$$X = \{x_1, x_2, x_3, x_4, x_5\}$$

Representation of elements in a given set

Consider the universe X with 5 elements,

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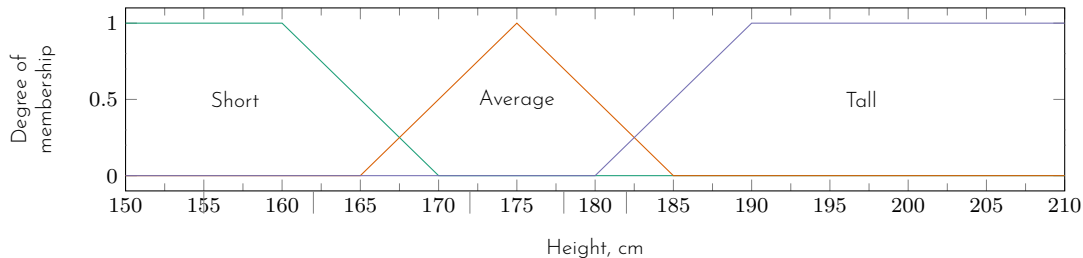
where $\mu_A(x_i)$ is the membership function of element x_i in subset A .

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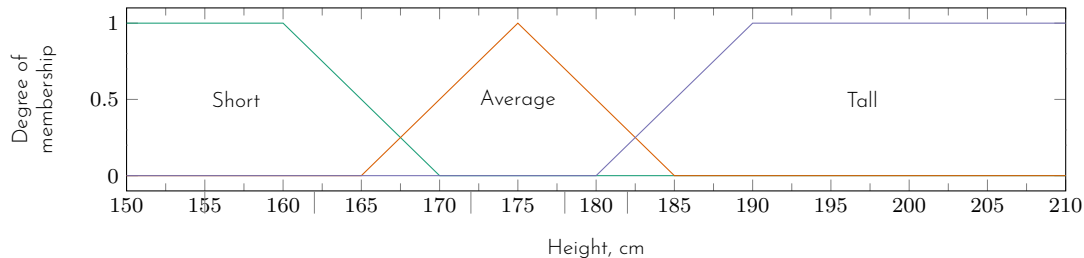
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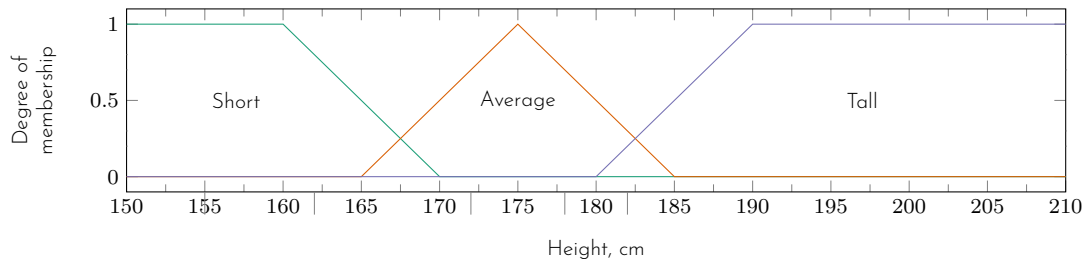
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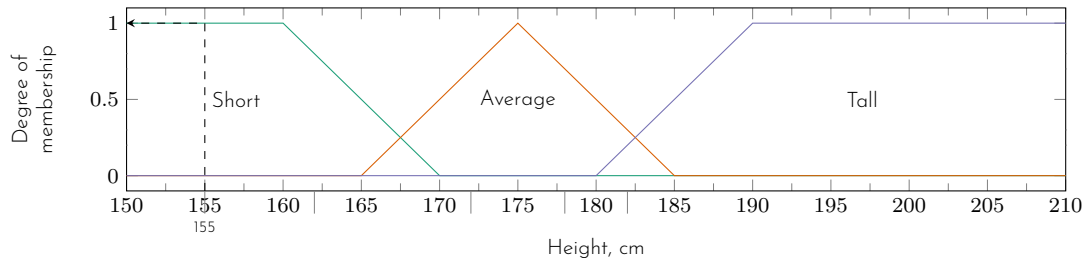


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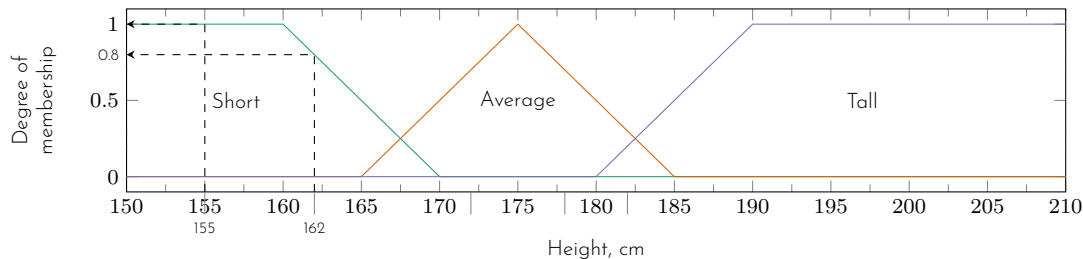


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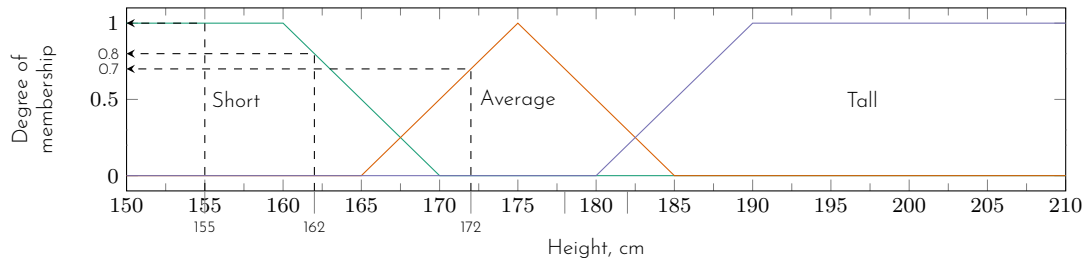


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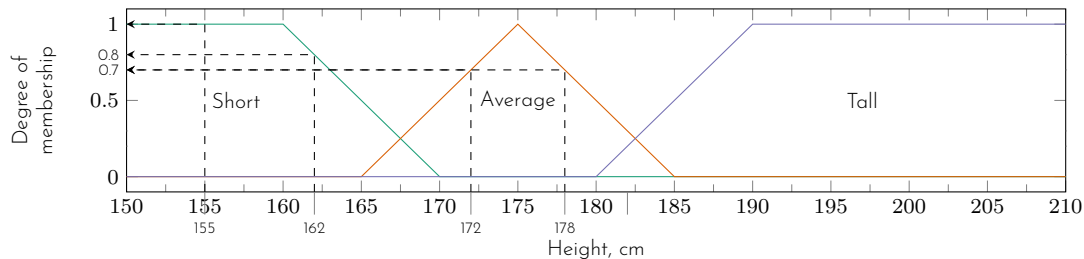


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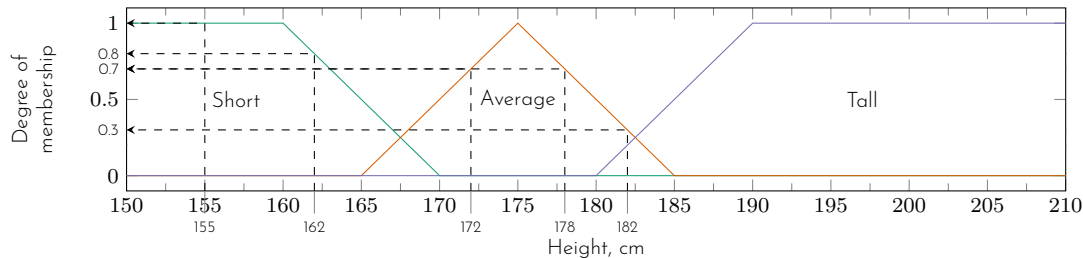


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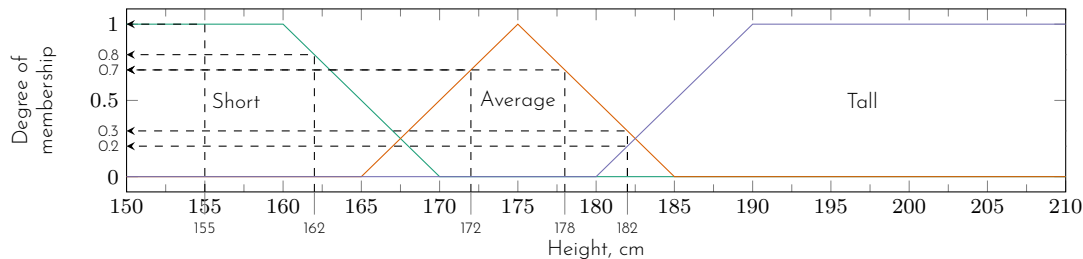


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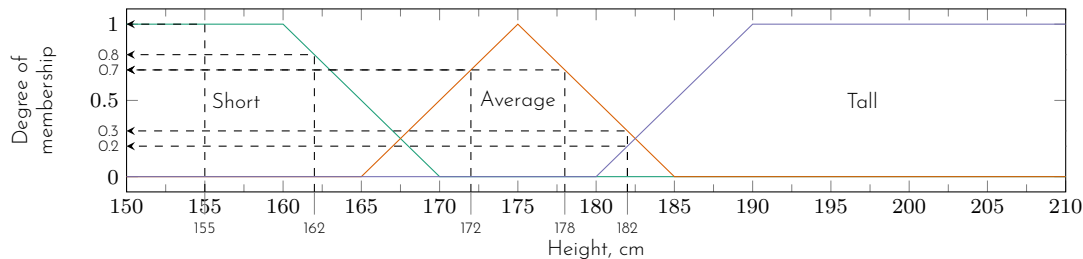


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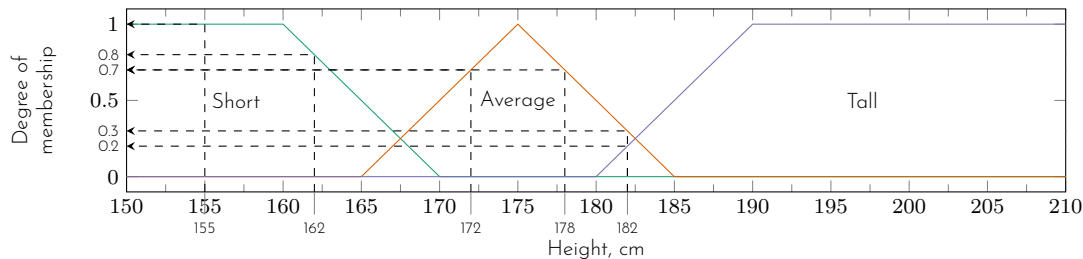
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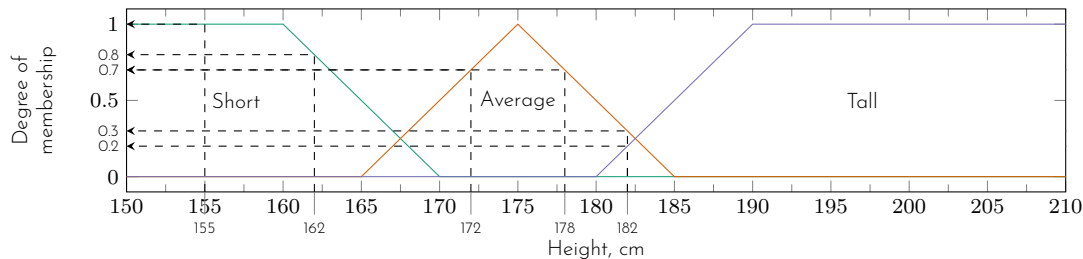
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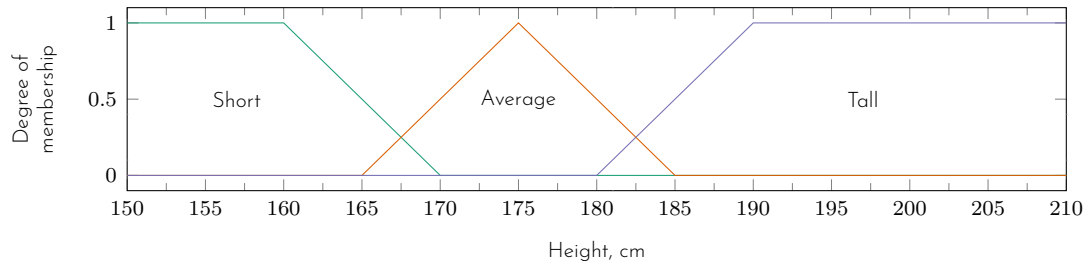
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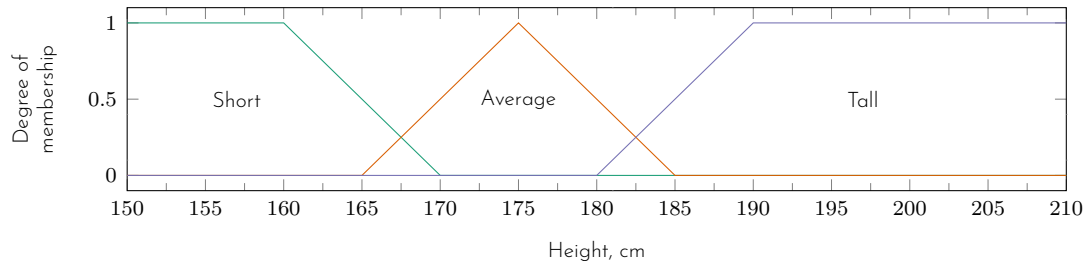
$$\text{tall men} = \{(155, 0), (162, 0), (172, 0), (178, 0), (182, .2)\}$$

Representation



How do we determine the degree of membership for a man with the height of 181cm?

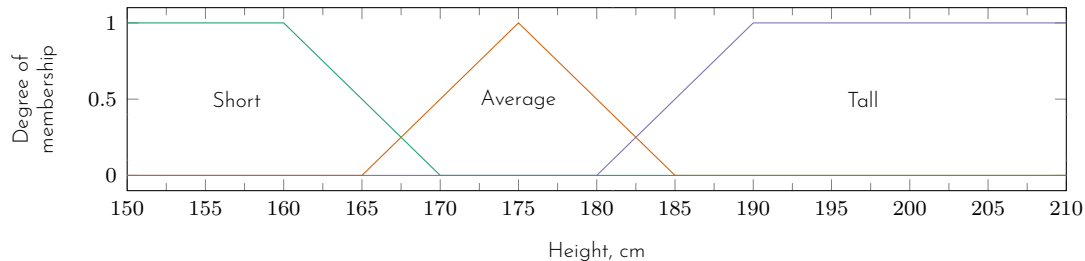
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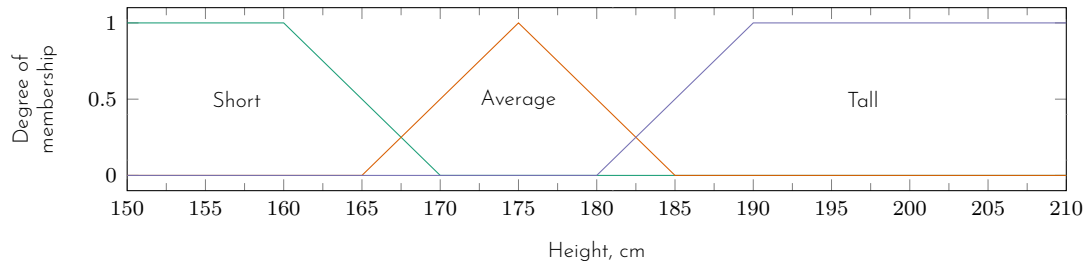
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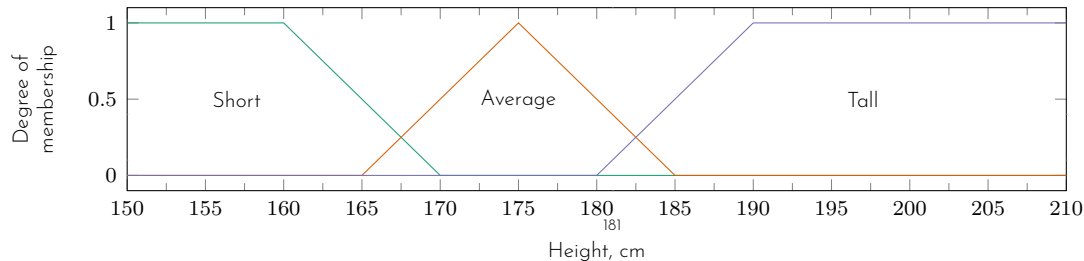
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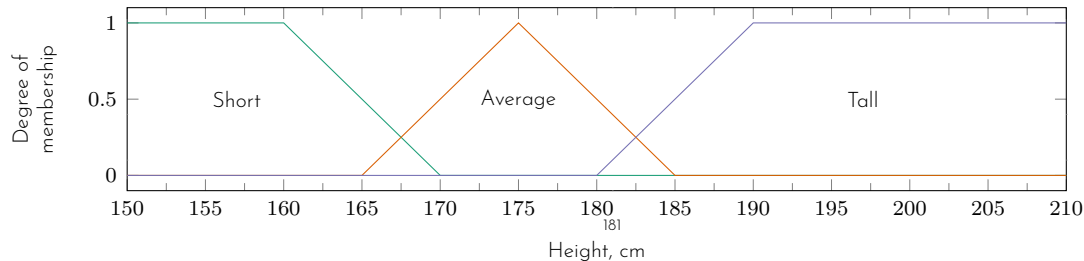
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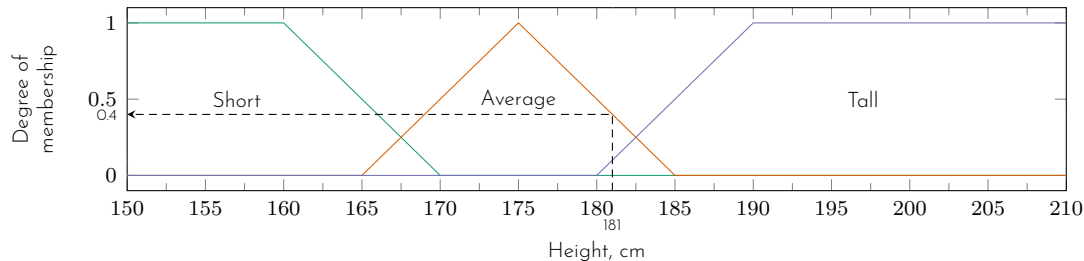
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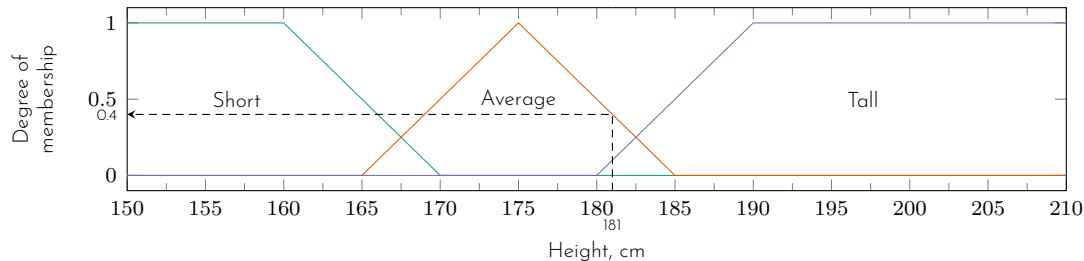
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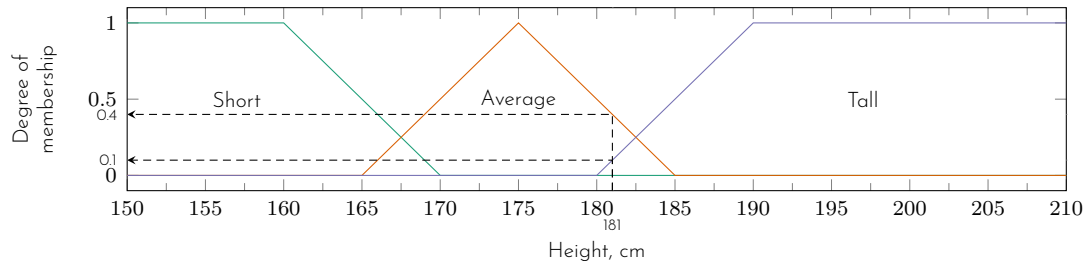
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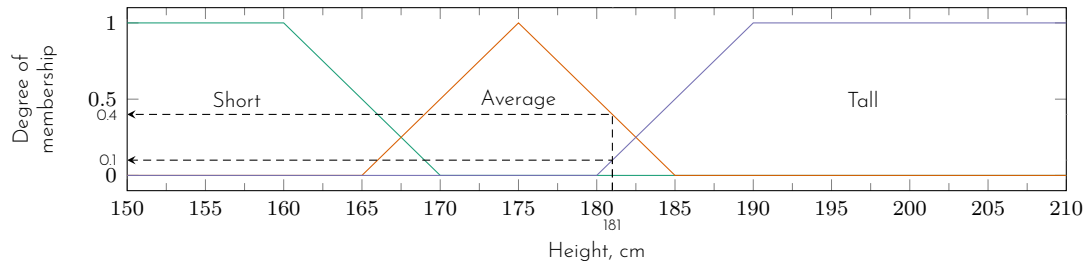
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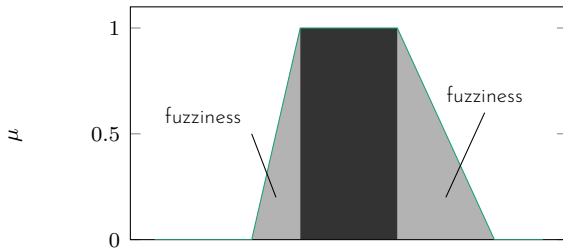
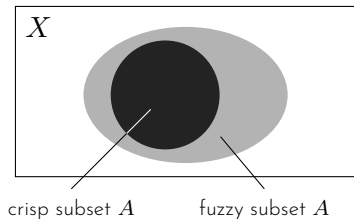
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Membership functions

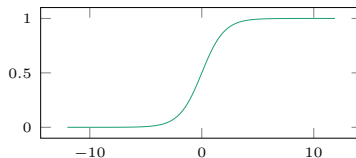
The difference between a crisp set and a fuzzy set is the introduction of fuzziness in membership function.



The typical membership functions used to represent a fuzzy set include

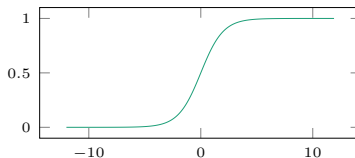
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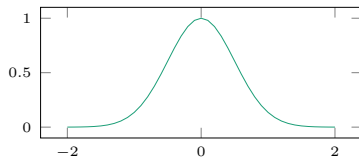


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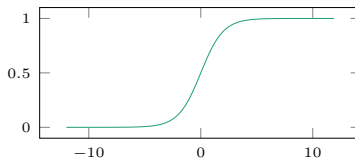
Gaussian



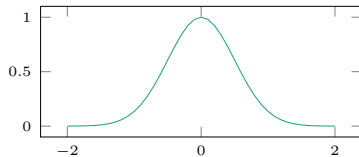
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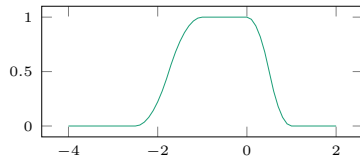
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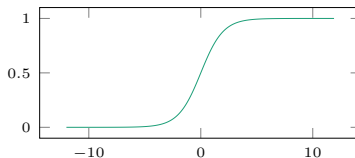
Pi



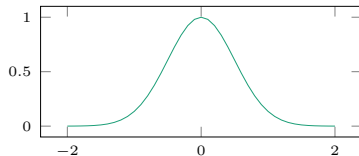
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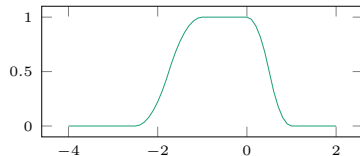
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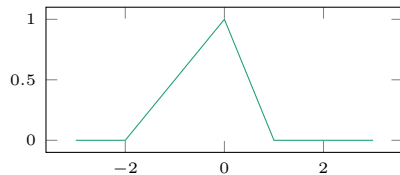


However, these functions increase the computation time.

Therefore, in practice, most applications use **linear fit functions**

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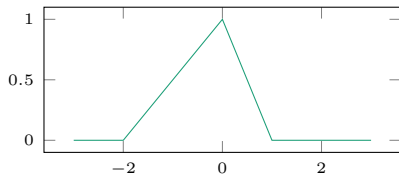
Triangular



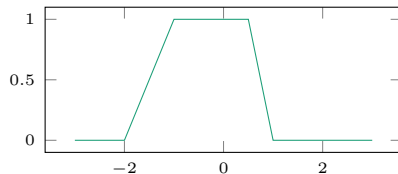
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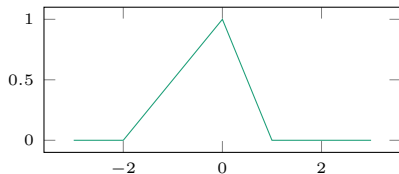
Trapezoidal



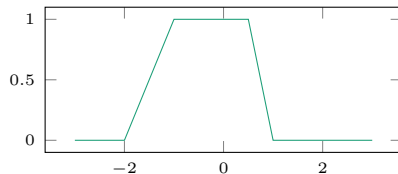
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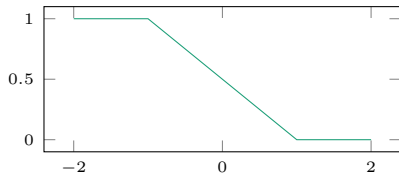
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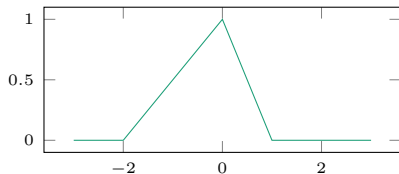


Z-curve

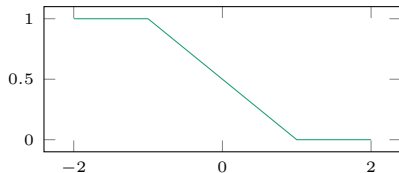


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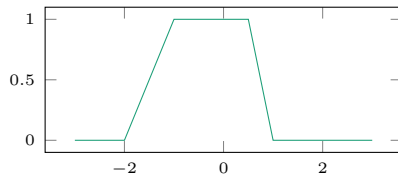
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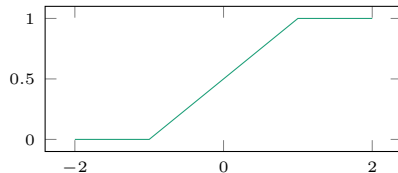
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S-curve

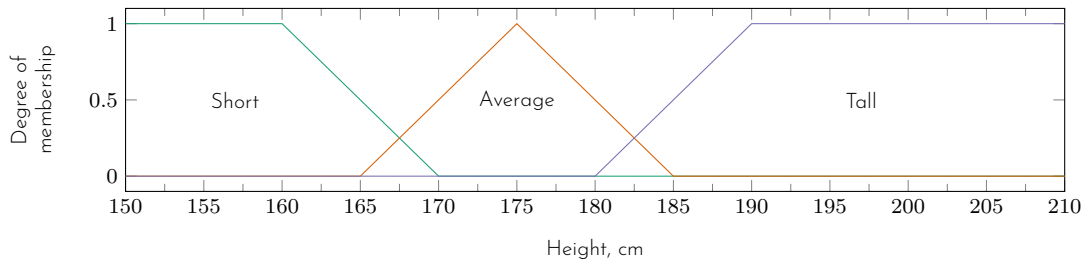


SUNWAY UNIVERSITY Jeffrey Cheah Foundation
Nurturing the Seeds of Wisdom

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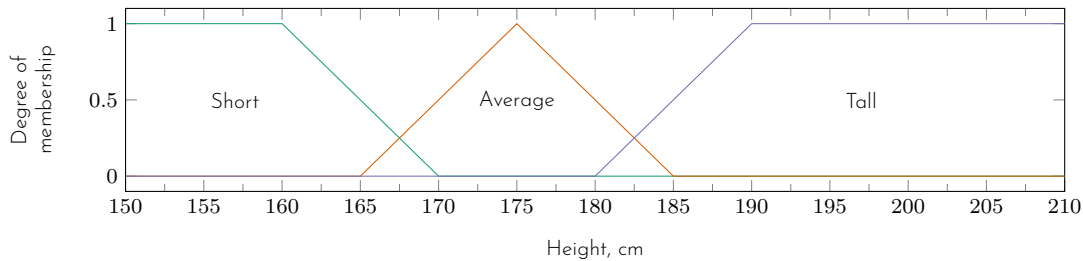
Representation of a fuzzy set

A fit vector is used to described each fuzzy set. Using the example of the fuzzy sets of men's height,



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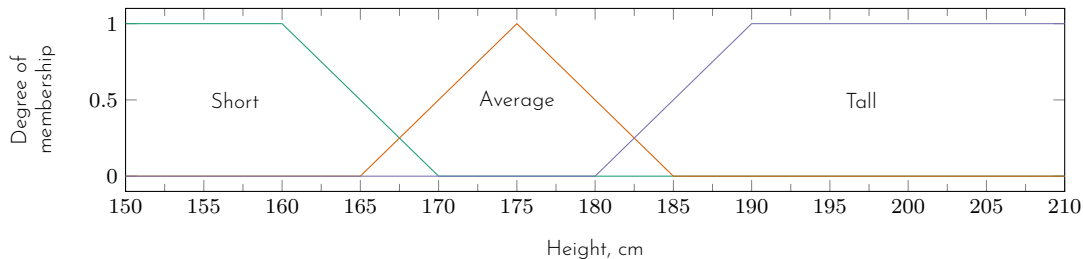
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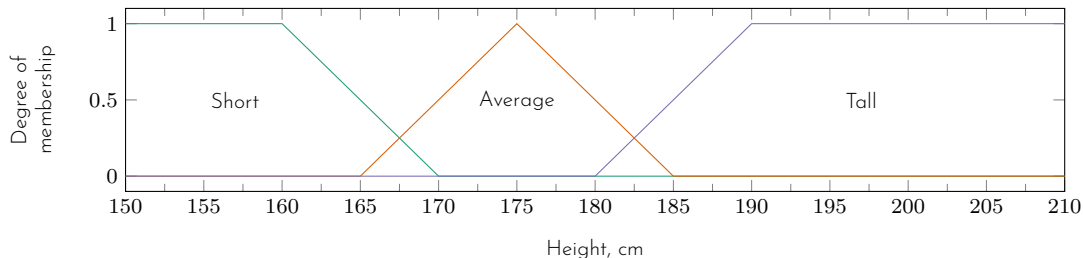


The fit vector for

$$\text{short men} = (1/160, 0/170) \text{ or } (1/150, 1/160, 0/170)$$

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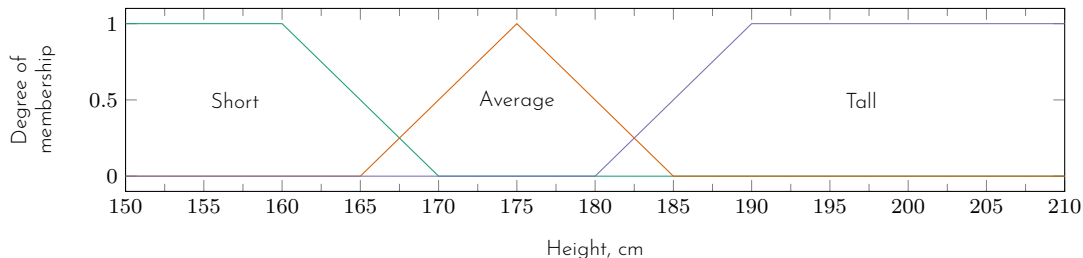
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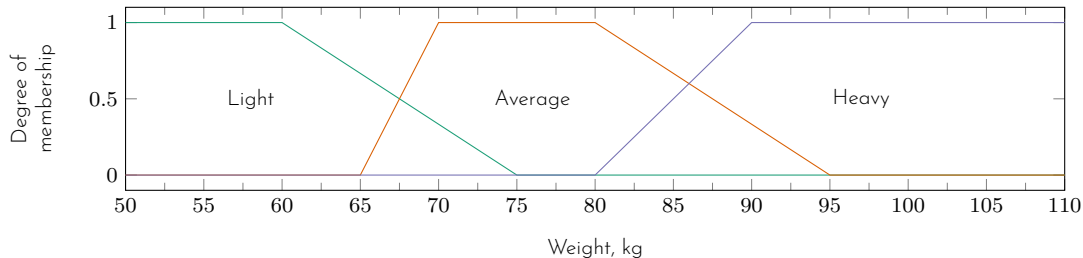
The fit vector for

short men = $(1/160, 0/170)$ or $(1/150, 1/160, 0/170)$

average men = $(0/165, 1/175, 0/185)$

tall men = $(0/180, 1/190)$ or $(0/180, 1/190, 1/210)$

Example 3



What is the fit vector for each fuzzy set of the fuzzy variable **weight**?

- ▶ A linguistic variable is a fuzzy variable.

Linguistic variables

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'John is tall'

linguistic variable: John

linguistic value: tall

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linguistic variable: John

linguistic value: tall

'wind is strong'

linguistic variable: wind

linguistic value: strong

- 'John is tall'

linguistic value: tall

'wind is strong'

linguistic value: strong

'speed is slow'

linguistic value: slow

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fuzzy subsets: very slow, slow, medium, fast, very fast

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Each fuzzy subset represents a linguistic value of the corresponding linguistic variable.

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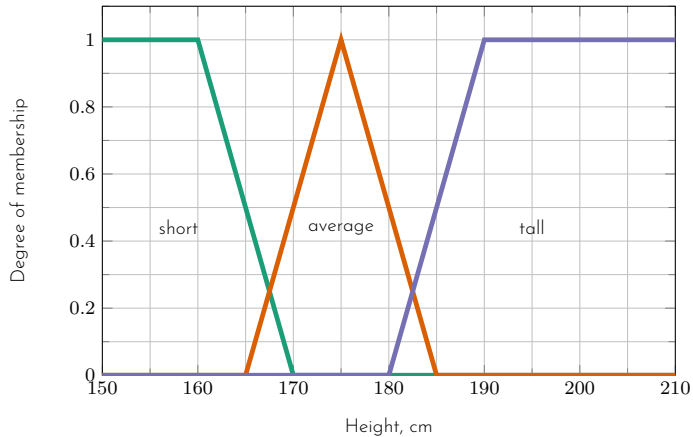
a little	slightly
very	extremely
very very	more or less
somewhat	indeed

- ▶ Hedges act as operations.

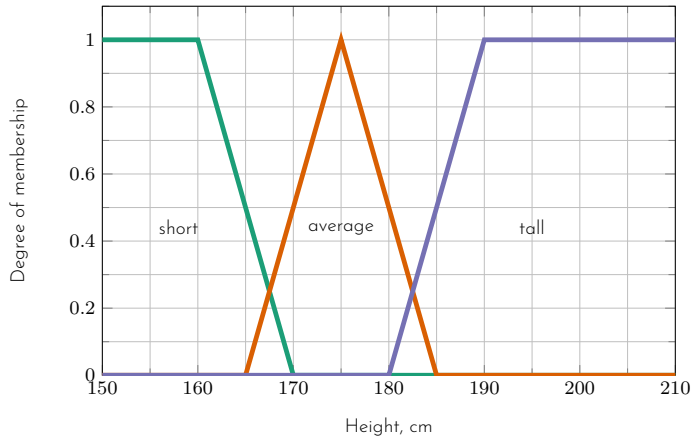
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- ▶ When a hedge is applied on a linguistic value, it changes the shape of the fuzzy set.

- ▶ Hedges act as operations.
- ▶ When a hedge is applied on a linguistic value, it changes the shape of the fuzzy set.
- ▶ For example, if 'very' is applied to a linguistic value A , it changes the shape from μ_A to $(\mu_A)^2$

Example 4

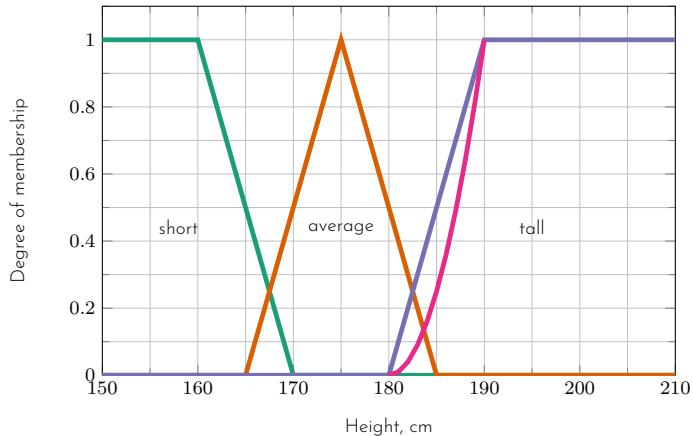


Example 4



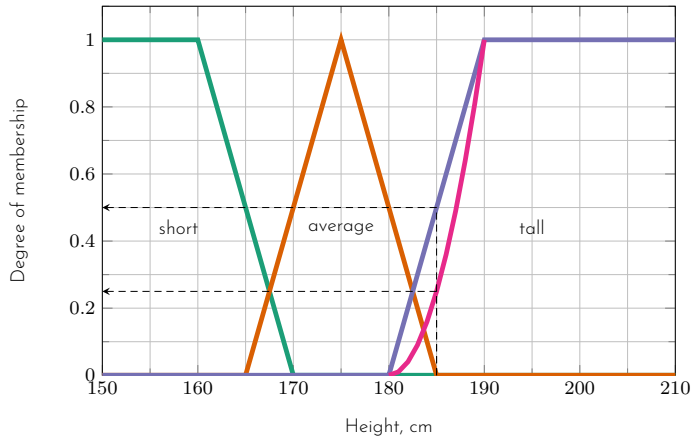
'very tall' (i.e. applying 'very' on the fuzzy set 'tall') $\rightarrow \mu_A^{\text{very}} = (\mu_A)^2$

Example 4



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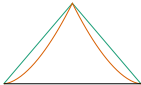
These are the hedges that are commonly used in practical applications:

Hedge	Mathematical expression	Graphical representation
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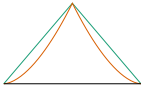
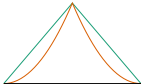
These are the hedges that are commonly used in practical applications:

Hedge	Mathematical expression	Graphical representation
a little	$(\mu_A)^{1.3}$	

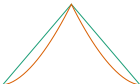
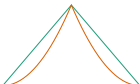
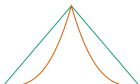
These are the hedges that are commonly used in practical applications:

Hedge	Mathematical expression	Graphical representation
a little	$(\mu_A)^{1.3}$	
slightly	$(\mu_A)^{1.7}$	

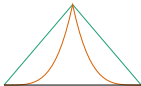
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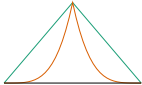
Hedge	Mathematical expression	Graphical representation
a little	$(\mu_A)^{1.3}$	
slightly	$(\mu_A)^{1.7}$	
very	$(\mu_A)^2$	

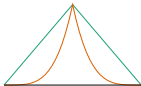
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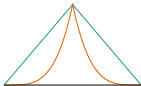
Hedge	Mathematical expression	Graphical representation
a little	$(\mu_A)^{1.3}$	
slightly	$(\mu_A)^{1.7}$	
very	$(\mu_A)^2$	
extremely	$(\mu_A)^3$	

Hedge	Mathematical expression	Graphical representation
Long call	$\Delta = N(d_1)$	
Long put	$\Delta = -N(-d_1)$	
Short call	$\Delta = -N(d_1)$	
Short put	$\Delta = N(-d_1)$	

Hedge	Mathematical expression	Graphical representation
very very	$(\mu_A)^4$	

Hedge	Mathematical expression	Graphical representation
very very	$(\mu_A)^4$	
more or less	$\sqrt{\mu_A}$	

Hedge	Mathematical expression	Graphical representation
very very	$(\mu_A)^4$	
more or less	$\sqrt{\mu_A}$	
somewhat	$\sqrt{\mu_A}$	

Hedge	Mathematical expression	Graphical representation
very very	$(\mu_A)^4$	
more or less	$\sqrt{\mu_A}$	
somewhat	$\sqrt{\mu_A}$	
indeed	$2(\mu_A)^2$ if $0 \leq \mu_A \leq 0.5$ $1 - 2(1 - \mu_A)^2$ if $0.5 < \mu_A \leq 1$	

Example 5

Using the fit vector for

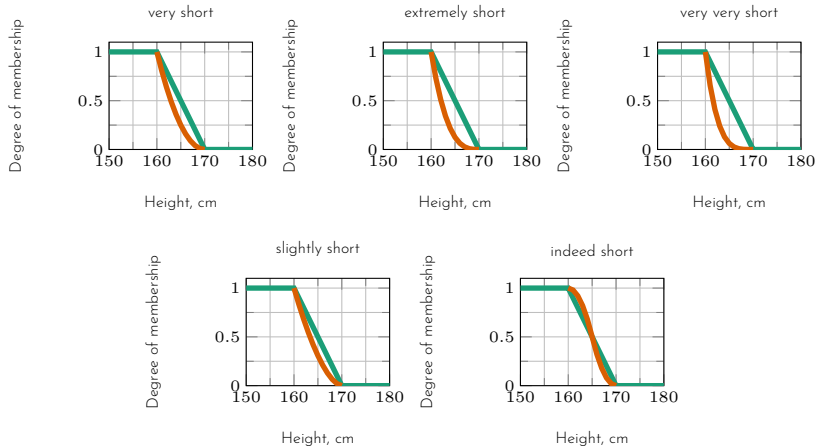
$$\text{short} = (1/160, 0/170),$$

show the membership values of the descriptions

- ▶ very short,
- ▶ extremely short,
- ▶ very very short,
- ▶ slightly short,
- ▶ indeed short.

on a fuzzy set diagram.

Example 5



Example 6

Given a fuzzy system of weights in kg with the fit vectors

$$\text{light} = (1/50, 0/60)$$

$$\text{average} = (0/55, 1/65, 1/70, 0/90)$$

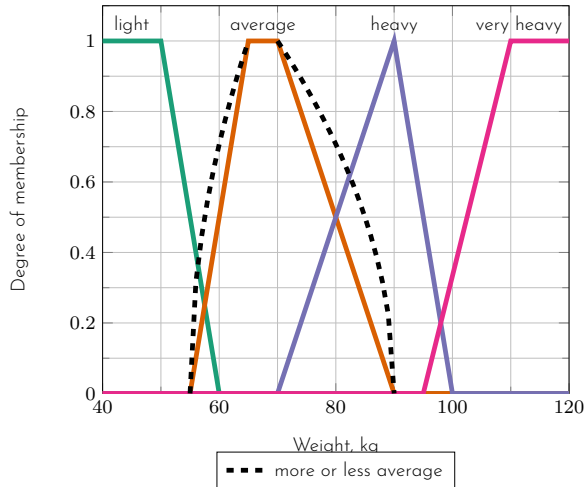
$$\text{heavy} = (0/70, 1/90, 0/100)$$

$$\text{very heavy} = (0/95, 1/110)$$

Draw the fuzzy set diagram.

Show the membership values of 'more or less average' on the fuzzy set diagram.

Example 6



Operations of fuzzy sets

Sets interact with one another through **operations**.

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- ▶ Equality

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- ▶ Complement (NOT)

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- ▶ Containment (subset)

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- ▶ Containment (subset)
- ▶ Intersection (AND)
- ▶ Union (OR)

Equality of two sets

Equality of two sets

► Crisp sets

- Sets containing the same members are equal.
- $\{x_1, x_2, x_3\} = \{x_1, x_2, x_3\}$
- $\{x_1, x_2, x_3\} \neq \{x_1, x_2, x_4\}$



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► Fuzzy sets

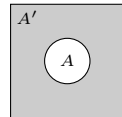
- Two sets are equal, if and only if, all elements have identical membership values.
- $\{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\} = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$
- $\{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\} \neq \{(x_1, 0.3), (x_2, 0.6), (x_3, 0.8)\}$

Complement (NOT)

Complement (NOT)

► Crisp sets

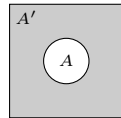
- All the elements that are not in the set (A' or $\bar{A}$)
- For $X = \{x_1, x_2, x_3, x_4\}$ and $A = \{x_1, x_3\}$,
 $A' = \{x_2, x_4\}$



Complement (NOT)

► Crisp sets

- All the elements that are not in the set (A' or $\bar{A}$)
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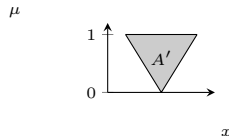
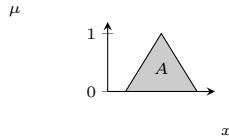


► Fuzzy sets

- Membership values of a complement set are given by

$$\mu_{A'} = 1 - \mu_A.$$

- If $A = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$,
then $A' = \{(x_1, 0.9), (x_2, 0.4), (x_3, 0.2)\}$

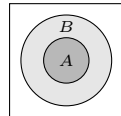


Containment (subset)

Containment (subset)

► Crisp sets

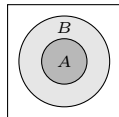
- $A \subset B$ if and only if all elements in A are also in B .



Containment (subset)

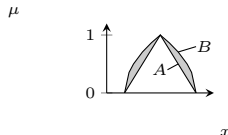
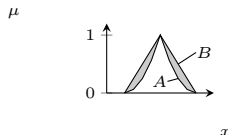
► Crisp sets

- $A \subset B$ if and only if all elements in A are also in B .



► Fuzzy sets

- $A \subset B$ if the membership values for each element in A is less than or equal to the membership value of the corresponding element in B .
- A hedge creates a subset or superset.
- If $A = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$ and $B = \{(x_1, 0.1), (x_2, 0.8), (x_3, 1.0)\}$, then $A \subset B$.

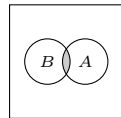


Intersection (AND)

Intersection (AND)

► Crisp sets

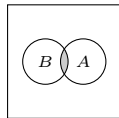
- The intersection of two sets contains those elements in both sets.
- If $A = \{x_1, x_2, x_3, x_4\}$ and $B = \{x_1, x_4, x_5, x_6\}$,
 $A \cap B = \{x_1, x_4\}$.



Intersection (AND)

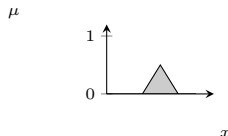
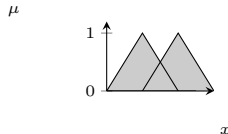
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► Fuzzy sets

- The weakest element determines the degree of membership in the intersection.
- $\mu_{A \cap B} = \mu_A \cap \mu_B = \min(\mu_A, \mu_B)$
- If $A = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$
and $B = \{(x_1, 0.5), (x_2, 0.3), (x_3, 1.0)\}$,
then $A \cap B = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.8)\}$.



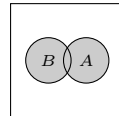
Operations of fuzzy sets

Union (OR)

Union (OR)

► Crisp sets

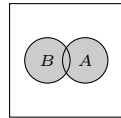
- All elements in either (or both) set(s) are included.
- If $A = \{x_1, x_2, x_3, x_4\}$ and $B = \{x_1, x_4, x_5, x_6\}$,
 $A \cup B = \{x_1, x_2, x_3, x_4, x_5, x_6\}$.



Union (OR)

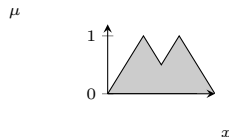
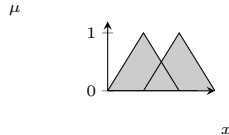
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► Fuzzy sets

- The strongest element determines the degree of membership in the union.
- $\mu_{A \cup B} = \mu_A \cup \mu_B = \max(\mu_A, \mu_B)$
- If $A = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$
and $B = \{(x_1, 0.5), (x_2, 0.3), (x_3, 1.0)\}$,
then $A \cup B = \{(x_1, 0.5), (x_2, 0.6), (x_3, 1.0)\}$.



Example 7

Given the universe of discourse,

$$X = \{x_1, x_2, x_3, x_4, x_5\}$$

and three fuzzy subsets,

$$A = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.8), (x_4, 0.6), (x_5, 1.0)\}$$

$$B = \{(x_1, 0.5), (x_2, 1.0), (x_3, 0.7), (x_4, 0.6), (x_5, 0.0)\}$$

$$C = \{(x_1, 0.3), (x_2, 0.1), (x_3, 0.2), (x_4, 0.8), (x_5, 0.4)\}$$

Find

- ▶ $A \cap B \cap C$
- ▶ $(A \cup B)'$
- ▶ $A \cap (B \cup C)$

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- ▶ $A \cap (B \cup C)$

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- ▶ $A \cap (B \cup C) = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.7), (x_4, 0.6), (x_5, 0.4)\}$

CSC3034 Current Course Update

02 Fuzzy Logic and Fuzzy Sets - presentation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current Lab 2 uses scikit-fuzzy to build and evaluate Mamdani control systems.
- The physical-AI extension applies fuzzy brake/throttle logic to a simulated robot.
- The updated solution checks optional dependencies and supports a standalone fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.